

# Creating a Windows Image

A golden image for the ALS image server

One reference machine, captured once, restored onto every unit that comes through. This is the procedure end to end - build, generalise, capture, publish, verify.

## Before you start: rebuild the audit stick's layer once

Restoring an image runs Clonezilla's `ocs-sr`, and the audit stick can only run programs that are inside it. `clonezilla` is now in the package list, but a stick built before that still lacks it and will stop at *"Clonezilla (ocs-sr) is not installed on this boot media"*. Run this once on the audit machine, then reboot:

```
sudo bash /cdrom/make-als-layer.sh build --with-session
```

The build prints *"ocs-sr baked in"* when it has worked. The separate Clonezilla USB used to CAPTURE images does not cover this — during a restore the machine is booted from the audit stick, and only one stick boots at a time.

## 1 Build the reference machine

Install Windows on a representative machine, then add drivers, every available Windows Update, and your standard software. Whatever is on this machine is what every customer receives.

Two things to get right before you go near Sysprep, because they cause the most common failure:

**Use a local account.** Do not sign in with a Microsoft account. Per-user Store apps are the number one cause of *"Sysprep was not able to validate your Windows installation"*.

**Pause Windows Update.** Left running, it re-provisions Store apps behind you and reintroduces the same failure after you have cleaned them out.

## 2 Sysprep — do not skip this

```
C:\Windows\System32\Sysprep\sysprep.exe /generalize /oobe /shutdown
```

**/generalize** strips the machine SID, hardware-specific drivers and the activation state. Without it every customer receives a clone carrying the same machine identity, which breaks domain joins and is visible to anyone who looks.

It shuts the machine down by itself. **Do not let it boot back into Windows** — if it does, you have to Sysprep again.

If it errors, the log names the offending app:

```
C:\Windows\System32\Sysprep\Panther\setupact.log
```

**Keep a pre-Sysprep snapshot.** `/generalize` can only run a limited number of times on one installation — three on retail licensing. Without a snapshot you cannot update the image later without rebuilding it from scratch.

### 3 Capture it to the server

Boot the reference machine from a **Clonezilla** USB — not the audit stick.

1. Choose **device-image**
2. Mount the server: **nfs\_server** for a Linux server, **samba\_server** for a Windows share
3. Point it at /srv/als-images (or //server/als-images)
4. Choose **savedisk**
5. Name it to match the manifest entry, e.g. win11pro
6. Accept the defaults for compression and splitting

**Use a cable.** 15 GB over gigabit ethernet takes about two and a half minutes. The same image over Wi-Fi on older hardware takes twenty to thirty.

### 4 Publish it

Add an entry to manifest.json on the server. The dir value must match the folder name exactly:

```
{ "id": "win11pro", "name": "Windows 11 Pro",  
  "version": "24H2 64-bit", "type": "clonezilla",  
  "dir": "win11pro", "icon": "windows" }
```

No code change is needed — the kiosk reads this file at startup. An entry whose folder is missing appears as **(image missing)** and cannot be selected, so a half-copied image can never be restored by accident.

### 5 Verify before you trust it

The restore checks for a clonezilla-img file inside the folder, so confirm it exists:

```
ls /srv/als-images/win11pro/clonezilla-img
```

Then restore it onto a scrap machine and boot it once, through to the Windows out-of-box setup. **An image nobody has restored is a guess**, and the moment to discover a bad capture is now, not when a customer's machine is half-written.

### Worth knowing

|                                |                                                                                                                                                                        |
|--------------------------------|------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| <b>Single point of failure</b> | If the image server is down, imaging stops on every bench. Keep one stick carrying a copy of the current Windows image as a fallback.                                  |
| <b>The share is read-only</b>  | Stations mount it read-only, so a compromised machine being imaged can never corrupt or encrypt your golden images.                                                    |
| <b>Server off or cable out</b> | The mount is soft, so a station does not hang. It warns and falls back to whatever images are on the stick.                                                            |
| <b>Serve time as well</b>      | Second-hand machines usually have a dead CMOS battery and boot with the wrong date, which breaks HTTPS and makes audits fail to upload. Run chrony on the same server. |

Full server setup — installing the share, the folder layout and pointing stations at it — is in `tools/IMAGE-SERVER-SETUP.md`.